

TemporalTemp: Encoding Historical Temperatures for Automated Sea Ice Mapping

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Abstract

We investigate the impact of incorporating **historical ERA5 reanalysis data** into deep learning models for Arctic sea ice classification. Unlike existing methods that rely solely on contemporaneous data, **we encode temporal variations in 2-meter air temperature ($t2m$)** to improve prediction of sea ice concentration (SIC), stage of development (SOD), and floe size (FLOE). Using a **multi-task U-Net** and the AI4Arctic dataset, we compare temporal encoding strategies to a baseline model without historical context. **A 1-year encoding with four 90-day samples yields the best performance** (85.62% combined score), with **SOD and FLOE accuracy gains of 1.43% and 1.20%**. While temporal context improves classification during seasonal transitions, it can degrade SIC estimates during melt periods, underscoring the need for complementary data sources.

Keywords: Sea Ice Mapping, Temperature Encoding, ERA5 Reanalysis, AI4Arctic, U-Net

Introduction

- Accurately mapping Arctic conditions is crucial for maritime safety, global climate regulation, and the well-being of Indigenous communities [1]-[3].
- Changes in Arctic sea ice are strongly influenced by melt and freeze-up timing [4].
- Air temperature is a strong predictor of these transitions [5].

Objective: Evaluate the use of historical ERA5 reanalysis 2-metre air temperature ($t2m$) data for improving sea ice parameter prediction, compared to Chen et al. [6].

Dataset

- The AI4Arctic dataset [7] consists of 533 scenes collected from Jan 2018 - Dec 2021.

Data Type	Feature	# of Channels
Sentinel-1	Dual-polarized (HH, HV) SAR images Gridded incidence angle measurements	2 1
AMSR2	18.7 GHz H, V and 36.5 GHz H, V brightness temperature	4
ERA5	Eastward, northward 10 m wind speed 2-metre air temperature Total column water vapor Total column cloud liquid water Historical 2-metre air temperature ($t2m$)	2 1 1 1 n
Auxiliary	Gridded latitude, longitude Acquisition month Distance-to-land map	2 1 1

Table 1:

Model input channels by data type. Adapted from Chen et al. [6], with n historical $t2m$ features added from the ERA5 reanalysis dataset.

A feature refers to the average of the $t2m$ timeseries over a time interval d .

Methodology

- To encode historical $t2m$ data, we apply a sampling strategy parameterized by the number of samples (n) and the length of each sample window (d , in days).

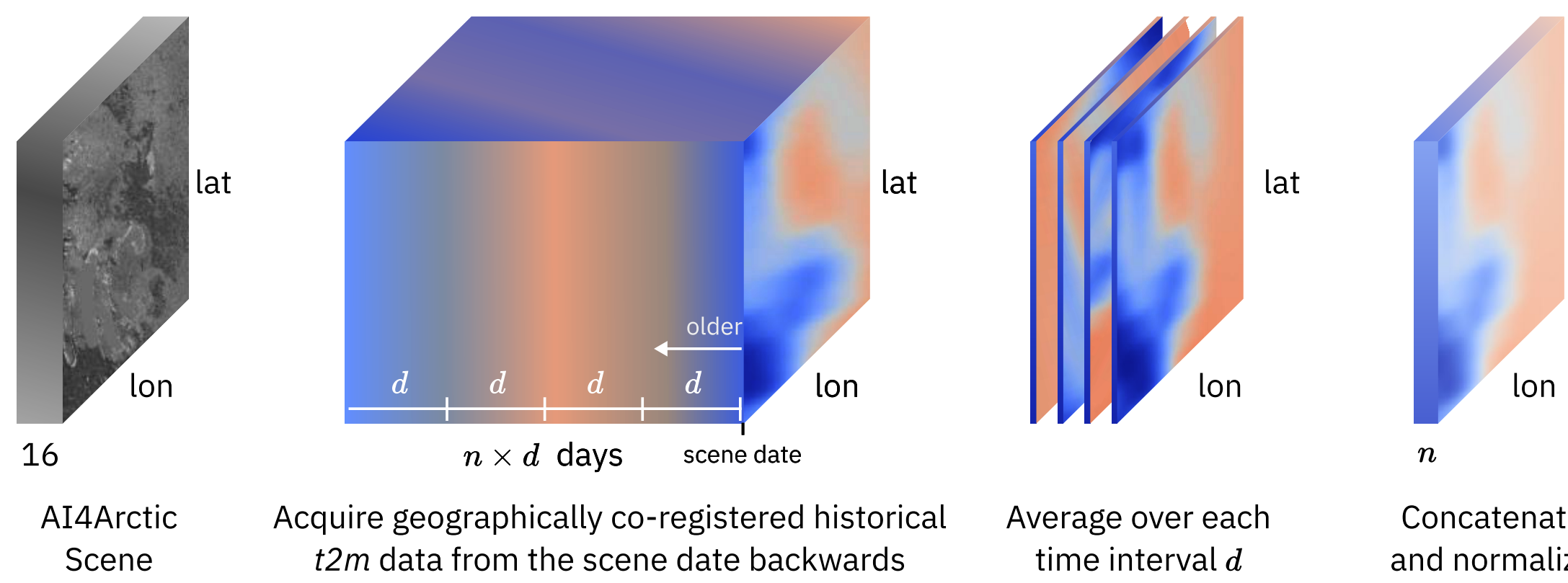


Figure 1: Co-registration and encoding of $t2m$ data for a single AI4Arctic scene.

- We use a multi-task U-Net architecture to simultaneously predict SIC, SOD, and FLOE.
- Encoded historical temperatures from Figure 1 are fused at the input-level with the 16 base channels detailed in Table 1.
- We use the same training and model configuration as Chen et al. [6] in our experiments.

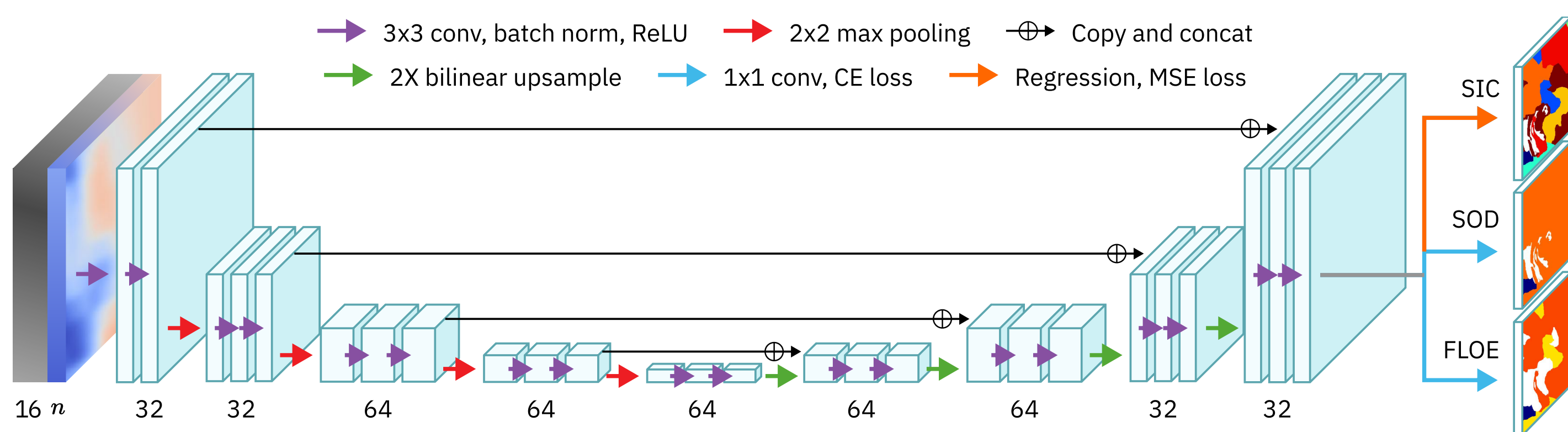


Figure 2: Multi-task U-Net architecture used in all experiments. The labels below each layer represent the number of channels, where n is the number of $t2m$ samples.

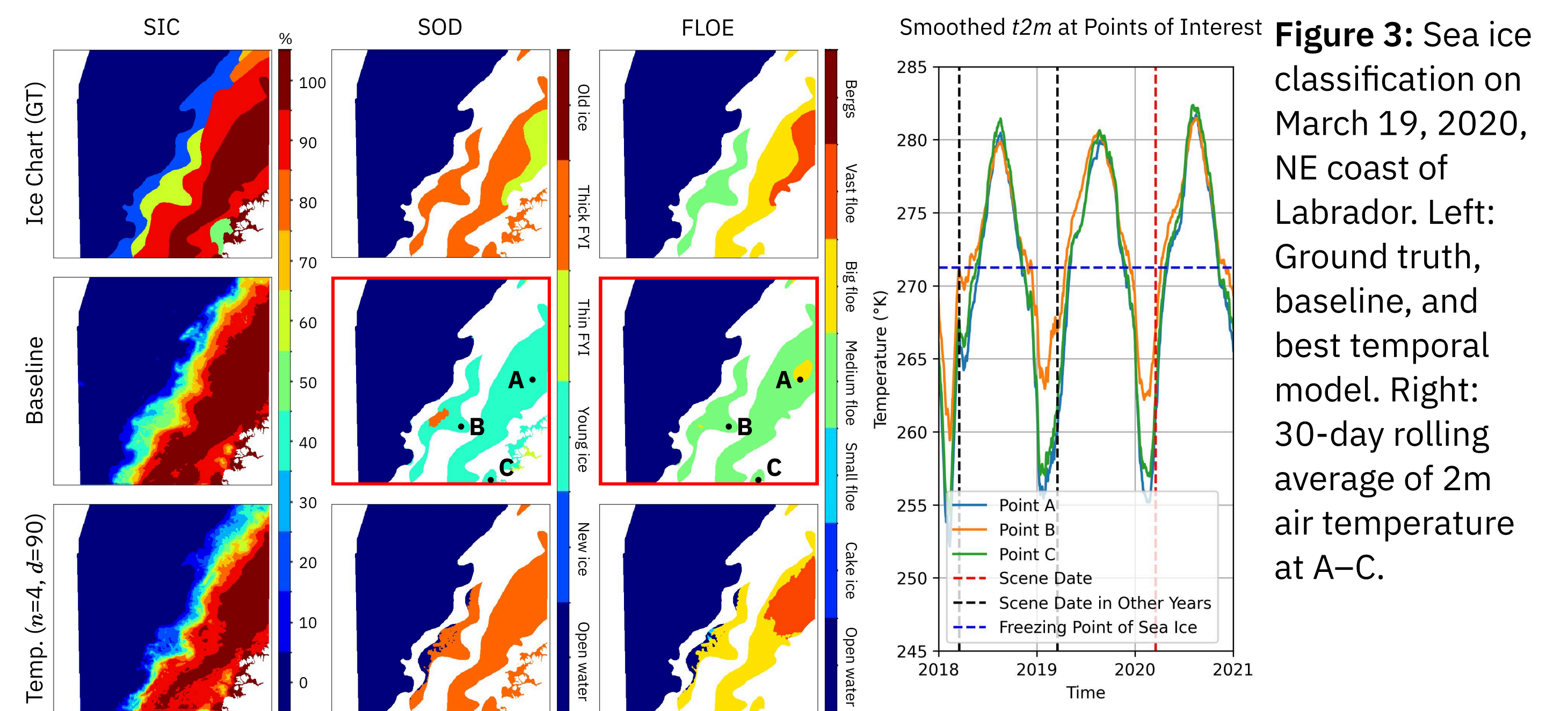
Results

Table 2: Scores of temporal encoding configurations across 20 test scenes. Metrics are mean $\pm$ standard deviation across 10 cross-validation runs. Baseline refers to Chen et al.'s [6] model.

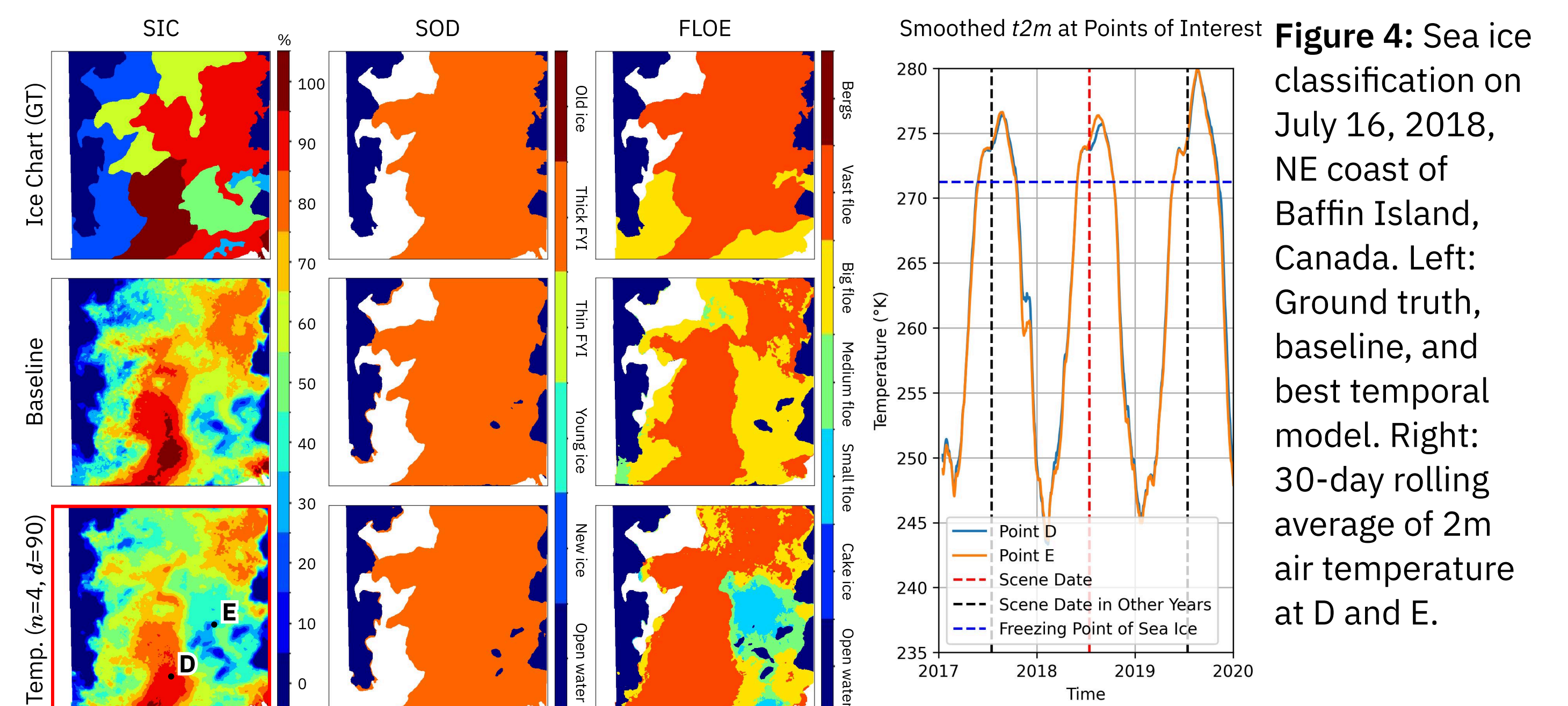
Encoding Length	$t2m$ samples n	Sample length d [Days]	Test Scores [%]			
			Combined	SIC R ²	SOD F1	FLOE F1
Baseline	0	-	85.00 $\pm$ 0.49	90.23 $\pm$ 0.35	86.10 $\pm$ 0.86	72.33 $\pm$ 1.14
3 Months	12	7	85.20 $\pm$ 0.76	89.78 $\pm$ 0.42	86.46 $\pm$ 1.54	73.54 $\pm$ 1.08
	24	3	84.40 $\pm$ 0.86	89.01 $\pm$ 0.54	85.75 $\pm$ 1.37	72.48 $\pm$ 2.53
4 Months	9	14	85.35 $\pm$ 1.06	89.82 $\pm$ 0.64	87.14 $\pm$ 1.62	72.86 $\pm$ 1.49
1 Year	2	180	84.81 $\pm$ 0.69	89.70 $\pm$ 0.55	85.83 $\pm$ 1.52	73.00 $\pm$ 1.08
	3	120	85.39 $\pm$ 1.62	89.61 $\pm$ 1.14	87.20 $\pm$ 2.63	73.31 $\pm$ 2.16
	4	90	85.62 $\pm$ 0.75	89.75 $\pm$ 0.71	87.53 $\pm$ 1.16	73.53 $\pm$ 1.10
	6	60	85.32 $\pm$ 0.94	89.32 $\pm$ 0.73	87.32 $\pm$ 1.52	73.33 $\pm$ 1.06
	12	30	85.14 $\pm$ 0.80	89.36 $\pm$ 0.64	86.95 $\pm$ 1.60	73.08 $\pm$ 1.11
	24	15	84.44 $\pm$ 0.82	88.86 $\pm$ 0.70	86.16 $\pm$ 1.51	72.17 $\pm$ 2.27

- ($n = 4$, $d = 90$) has the best combined score, improving in SOD and FLOE over the baseline by 1.43% and 1.20% respectively.
- Longer encodings improve performance by better representing temperature trends.
- Small n fails to capture sufficient detail, large n causes overfitting on minor fluctuations.

Discussion and Conclusion



- At points A-C (boxed in red), the baseline model mislabels thick first-year ice as young ice (SOD) and misclassifies large floes as medium floes (FLOE).
- In 2018, $t2m$ at point B exceeded the sea ice melting point, while 2019 and 2020 remained colder. Melt onset in 2020 was slightly later than 2019.
- Insight:** Encoding $t2m$ directly, rather than using calendar-based features, captures interannual variability in freeze and melt timing. This makes the model more adaptable to changing environmental conditions.



- At points D and E (boxed in red), the best temporal model underestimates SIC.
- Despite the presence of consolidated ice, $t2m$ at the scene date exceeded the freezing point of sea ice. This suggests that surface melt likely decoupled atmospheric temperature from underlying ice, reducing $t2m$'s reliability as a melt indicator.
- Insight:** Encoding $t2m$ can degrade performance when air temperatures rise without corresponding ice loss, reinforcing misleading correlations.

Conclusion

- Combining historical air temperature data with SAR and passive microwave data can improve automated sea ice classification, particularly for SOD and FLOE during transitional seasons.
- Future work should explore adaptive weighting mechanisms to reduce the influence of temperature features when they conflict with other sensor data.

References

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